Interaction with **Team Flyby**

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Introductions

Team Name

SINGULARIT

Composition

30 Engineering undergraduates from Cochin University of Science and Technology, India.

Previous Experience

Placed 25th out of 49 teams in SUAS 2025.

Current Projects

Attempting the Mathworks Minidrone Competition and A2RL.

Technical Difficulties

We would love to hear about the technical difficulties your team faced during development.

What felt different about this competition compared to others?

Technical Doubts and Questions

While preparing and designing the architecture for this project, we encountered some doubts and questions that we would like to ask your team.

Q. Were there any computation bottlenecks while using NVIDIA Jetson when the drone was at top speeds? What were the sensor modalities you are given to work with (anything other than CMOS camera & IMU)?

Q. Can you tell us more about your approach on perception using available sensor modalities?

Are you relying entirely on EKF & PnP for correcting state estimation (from drift) or using any end-to-end learned perception?

Also for localisation of state vector or computing observation vector, how do you get these known map ie. gate coordinates beforehand and how do you manage to do that?

Your approach on gate segmentation to get the corners, are you using any CNN based neural nets there to segment gate & extract these gate corner? If so any image processing like adaptive cropping is done before feeding it to neural net better retrieval?

Q. The indoor competition hall presented severe visual challenges, such as poor lighting, low contrast, and distracting background machinery, similar gates, motion blur, that caused gate misclassifications for other teams. How did you prepare your perception models for this specific environment without prior access ?

Did you rely on heavy synthetic data augmentation (like StochGAN) for testing & simulation of gate segmentation model, or did you fine-tune your models using logs from early practice flights in the hall?

Q. The standard Arducam IMX219 camera exhibits severe rolling shutter shearing during high-speed rotations. How was the team able to get around this?

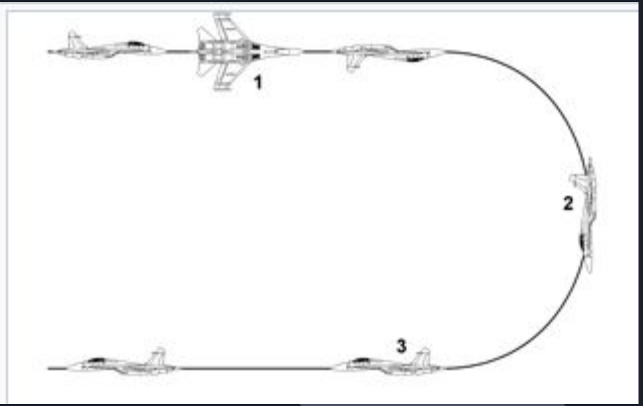
Q. Since the flexible TPU camera mounts on the A2RL drones were prone to shifting after crashes or physical repairs, how did you handle extrinsic calibration drift? Did you have to manually re-calibrate the camera between heats, or Did you rely on an offline IoU-based optimization between predicted gate reprojections and observed segmentation masks, or attempt any online calibration during flight?

how sensitive was your state estimator to small calibration residuals (e.g., $<1°$ error)?

How did your EKF handle the transition period after a crash when calibration was incorrect? Did you explicitly gate vision updates or inflate covariance to prevent estimator divergence?

Q. How did the team handle 16g IMU saturation fallback logic during aggressive maneuvers like the “Split-S”?

how did your state estimator handle accelerometer clipping? Did you implement dynamic model-based predictions as a fallback for the EKF, or did you dynamically inflate your covariance to rely entirely on vision?"



Split S Maneuvers

Q. How was collision detection implemented in the Silver Multi-Drone Race? Did you integrate opponent detection into your vision pipeline or is that something to do with your control policy?

Other teams struggled in this event because their AI lacked collision avoidance and crashed into opponents.

Q. Tell us more about your planning & control architecture and latency issues you faced there?

Did you generate time-optimal geometric paths tracked by an MPC, or did you use an RL policy to map your state estimates directly to CTBR commands?

Was there ever a need to force certain geometric flight paths along with RL policy or aggressive maneuvers like for multi-gate setups like Split-S or inverted loop?

For certain situations such as multiple stacked gates, a mathematically planned cardioid trajectory is the most optimal, was there a need to hard code these paths as polynomial trajectories by MPC?

Q. We know the standard A2RL drone compute relied entirely on the NVIDIA Jetson Orin NX. Did you use a betaflight fork for low-level inner loop firmware for STM32 flight controller and why (for reducing latency)?

We read teams like Monorace directly map state estimates to low level RPMs commands using their RL policy replacing CTBR outputs to the controller firmware like betaflight, how effective is that, did you try?

how did you balance the computational load of your perception, state estimation, and high-level control policies on the Jetson? Did you face any latency bottlenecks at high speeds?

Q. In multi-lap events, battery voltage sag significantly reduces the maximum motor RPM over time. Did your control policy assume a constant maximum thrust, or did your agent actively estimate the maximum attainable RPM dynamically to adapt its flight path and prevent crashing on later laps?

Q. Were there any software parameter adjustments to be made while shifting from simulation to real life testing? How you adapt RL policy to work in real env?

How different was testing in the real world compared to the simulated runs? Were there drastic differences, that had to be taken into account?

Competition related Queries

We would love to hear about your experience during the competition

Q1. A summary of the full selection process — initial application, followed by simulation rounds or qualification stages, before the team was confirmed for the live event?

We primarily want to know about the competition timeline, qualification dates and deadlines to submit a registration.

Q2. What was your initial video submission based on — was it footage from a drone you had already built? What elements did it contain?

We are currently editing our own video, taken from our mission run at SUAS. Do you have any suggestions on how to make the video?

Q3. Was there any financial support provided by the organizers? In the form of hotel rooms or flights?

Q5. Was the whole experience enriching for the team? How was the reception at the venue and were there opportunities to network and collaborate?

Do you have any questions for us?

Thank you!

For taking time out of your busy schedule for us, we are truly grateful for Team Flyby!